

Modification and Improvement of a Coffee Berry Sorting Machine

Continuous Assessment Presentation — FYP 22-22

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March 2026

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Introduction & Background

Context:

- Coffee is Kenya's leading export crop; bean quality is determined at sorting stage.
- Manual sorting is labour-intensive and inconsistent â€” ripe (deep-red) berries must be separated from under-ripe, over-ripe, and defective ones.
- A prototype automated sorter was developed by a previous group using a Raspberry Pi 3B+, Pi Camera V2, and a 4-class MobileNetV2 classifier.
- The existing machine achieves only $\approx 72\%$ accuracy at ≈ 35 berries/min â€” commercially inefficient



Visual diversity at harvest â€” ripe, semi-ripe, over-ripe, defective



Problem Statement

Deficiencies identified in the existing machine:

Aspect	Existing	Industry Target
Accuracy	$\approx 72\%$	$> 90\%$
Latency	≈ 380 ms	< 100 ms
Throughput	≈ 35 /min	> 150 /min
Camera	Rolling-shutter	Global-shutter
Driver	BJT (storage delay)	MOSFET (fast)
Classifier	4-class MobileNetV2	Binary YOLOv11n

Root causes:

- Slow TFLite inference on Pi 3B+ (310 ms of total delay)
- Rolling-shutter smear at 0.5 m/s chute

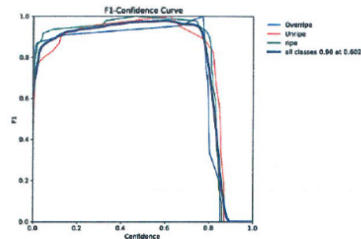


Figure 4.13: F1 score

Previous MobileNetV2 F1 score $\hat{=}$ "ripe/semi-ripe boundary confusion evident"

Objectives

Main Objective

To modify and improve an existing automated coffee berry sorting machine to achieve industry-grade accuracy, speed, and reliability.

Specific Objectives:

1. **Hardware upgrade:** Replace the rolling-shutter camera with a global-shutter module and replace BJT ejector drivers with logic-level MOSFET drivers; upgrade from Raspberry Pi 3B+ to Pi 4 (4 GB).
2. **AI model development:** Retrain a binary YOLOv11n classifier on a 2 000-image annotated dataset sourced from Roboflow, achieving $mAP@0.5 > 90\%$.
3. **System integration & validation:** Integrate the upgraded subsystems and validate end-to-end latency (< 100 ms), throughput (> 150 berries/min), and classification accuracy ($> 90\%$).

Literature Review – Key Sources

Author(s) / Source	Key Contribution	Limitation / Gap
Roychowdhury et al. (2023)	YOLOv5-based real-time fruit grading on conveyor at >95% mAP@0.5	Tested on static belt, no in-motion blur analysis
Tabiolo (2024) – Roboflow	Annotated 600-image coffee cherry dataset (accept/reject classes)	Small dataset; fixed controlled lighting only
Abdulridha et al. (2022)	CNN pipeline for coffee bean defect detection, 87% accuracy	Post-harvest beans only; no real-time sorting
Ononiwu et al. (2021)	MOSFET solenoid driver for high-speed pneumatic sorting	No Pi/microcontroller integration study
Ultralytics (2024)	YOLOv11n architecture: 2.6M params, 640×640 input, ONNX export	No domain-specific benchmarks for berry classification
Arducam (2023)	OV9281 global-shutter module eliminates rolling-shutter artefacts	Integration with Pi 4 + MIDI CSI requires custom

Gaps Identified & Proposed Approach

Gaps in existing work:

- No study combines global-shutter vision, MOSFET drivers, and a binary YOLO model on a Pi 4 platform for berry sorting.
- Existing datasets are small ($<1\,000$ images) and single-lighting-condition.
- No edge-deployment + co-processor (ESP32) latency study for pneumatic ejection in this context.
- 4-class models consistently show semi-ripe/ripe confusion in in-motion scenarios.

Our proposed approach:

1. **Binary classification:** collapse 4 classes to accept / reject " eliminates boundary confusion.
2. **Global-shutter camera:** Arducam OV9281 " eliminates motion blur.
3. **MOSFET driver:** IRLZ44N, 3.3 V logic-level " eliminates storage-time delay.
4. **Co-processing:** Pi 4 $\xrightarrow{\text{UART}}$ ESP32 " offloads real-time actuation from Linux scheduler.

5. **Large augmented dataset:** 5

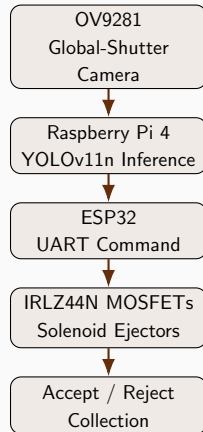
Methodology â€™ Design Approach

4-module improvement strategy:

#	Module	Improvement
1	Vision System	Global-shutter camera + binary YOLOv11n
2	Electronics	IRLZ44N MOSFET drivers; Pi 4 upgrade
3	Mechanical	CAD-redesigned chute, feeder, hopper
4	Software	Pi4 → UART → ESP32 pipeline

Design flow:

1. Baseline audit of existing machine
2. Targeted component-level redesign



End-to-end detection-to-actuation
pipeline

Methodology â€™ Vision Subsystem: Before & After

Before â€™ Existing

Camera	Pi Camera V2 (rolling shutter)
Classifier	MobileNetV2 TFLite
Classes	4 (ripe/semi/unripe/defect)
Accuracy	$\approx 72\%$
Inference	≈ 310 ms on Pi 3B+



After â€™ Proposed

Camera	Arducam OV9281 (global shutter)
Classifier	YOLOv11n (COCO pre-trained)
Classes	2 (accept / reject)
Target Acc.	$> 90\%$ mAP@0.5
Inference	Pi 4 (target < 100 ms)

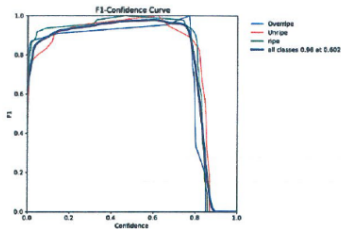


Figure 4.12: F1 score



Methodology â€™ Driver Electronics: Before & After

Before â€™ Existing

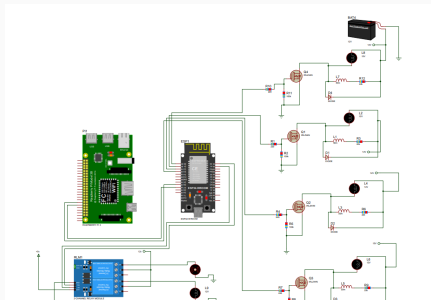
Driver transistor	PN2222A NPN BJT
Logic voltage	3.3 V GPIO (insufficient V_{BE})
Storage delay	≈ 500 ns
Gate resistor	N/A (base current via 1 k Ω)
Flyback diode	Not verified $\Rightarrow$

Problems:

- Insufficient base overdrive from 3.3 V GPIO
- Storage time causes solenoid mis-timing
- Risk of inductive flyback without guaranteed clamping

After â€™ Proposed

Driver transistor	IRLZ44N Logic-Level MOSFET
Logic voltage	3.3 V GPIO (full enhancement)
Gate resistor	220 Ω (no ringing)
Pull-down	100 k Ω (no false fire)
Flyback diode	1N5819 (clamps to 12.5 V)



Methodology â€™ Mechanical Structure: Before & After

Before â€™ Existing Assembly



Figure 3.16: Fabricated mechanical assembly

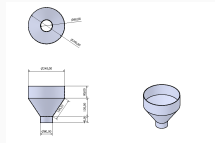
Fabricated machine â€™ previous group

Issues:

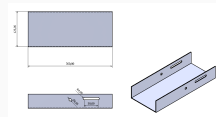
- Hopper geometry caused berry jamming
- Chute incline not optimised for



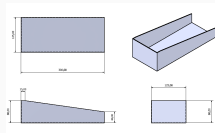
After â€™ CAD Redesigns



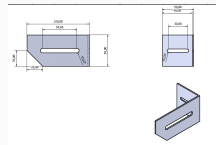
Redesigned hopper



Optimised chute



Feeder trough



Adjustable bracket

All components modelled in SolidWorks

Methodology â€” Dataset & YOLOv11n Training

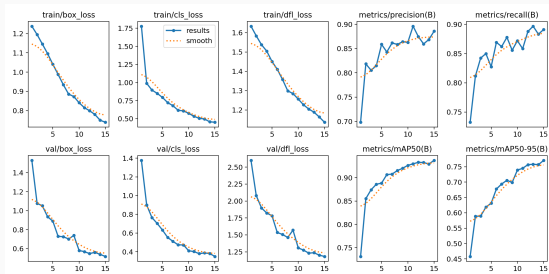
Dataset (2 000 images, 5 Roboflow

Roboflow Dataset		Images
Coffee	Cherry	563
Ripeness		
Coffee	Cherry	600
(Tabiolo)		
sources): Coffee App Classifi-		291
cation		
Cherry	Classifica-	298
tion		
Coffee	Berry De-	248
tection		

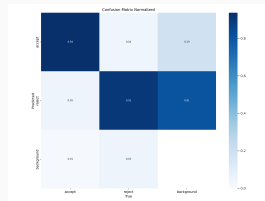


YOLOv11n validation batch â€”
accept/reject predictions

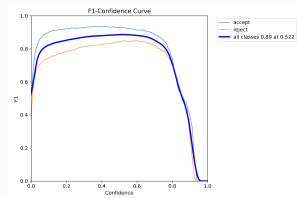
Results â€™ YOLOv11n Model Performance



Training curves: box loss, cls loss, precision, recall, mAP@0.5 over 15 epochs



Normalised confusion matrix



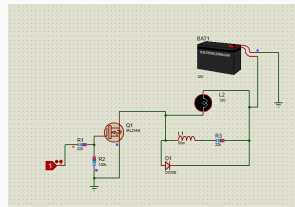
F1-confidence curve (binary: accept / reject)

Results â€™ Proteus Circuit Simulation

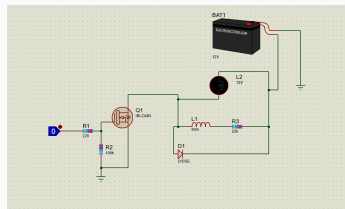
Verified in Proteus ISIS before hardware build:

- All 4 MOSFET channels fired correctly on REJECT:n commands
- UART parsed at 115 200 bps; simulated latency <10 ms
- No gate ringing â€™ 220 Ω gate resistor confirmed effective
- Flyback diode clamps inductive spike to 12.5 V (0.5 V above rail)
- 100 k Ω pull-down: zero false firing at power-on reset

MOSFET switching waveforms:



Turn-ON: gate 3.3 V, drain falls to ≈ 0 V
(solenoid fires)



Turn-OFF: 1N5819 flyback diode clamps

Budget

Component	Quantity	Price
Raspberry Pi 3	1	5500
Arduino 2560 Micro controller	1	3500
Camera	1	1500
Power Supply	1	3000
Air Ejectors	3	2500
Vibrating Motor	12v 200W (1)	1500
Relay	1	100

Conclusion & Way Forward

Summary of progress:

- Baseline deficiencies fully characterised
- CAD redesigns completed (hopper, chute, feeder, bracket)
- All 5 Roboflow datasets integrated; 2 000-image dataset assembled
- YOLOv11n trained for 15 epochs; training metrics collected
- Proteus circuit simulation verified â€” all MOSFET channels fired correctly
- Hardware components procured and bench-tested

Objective Progress:

Objective	Status	%
Hardware upgrade	In progress	60%
AI model	In progress	50%
Integration/valid.	Planned	0%

Way Forward:

Time Plan

Activity	1	2	3	4	5	6	7	8	9	10	11	12	13	14	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Concept Note																												
Literature Review																												
Project Proposal																												
Presentation																												
System Requirements Specifications																												
Mechanical design																												
Electrical design																												
Control system design																												
Mechanical body fabrication																												
Electrical wiring																												
Camera Integration																												
Final assembly																												
System Testing																												

Thank you

Questions?